A formula for D(-1)-quadruples

Problèmes ouverts — A. Dujella · 2026

Problem 1.3 (Dujella, 2026). Let $\{a, b, c\}$ be a $D(-1)$-triple and d a positive integer such that $ad + 1$, $bd + 1$, and $cd + 1$ are perfect squares. Do all such quadruples satisfy the formula

$$(d + a + b - c)^2 = 4(ab - 1)(cd + 1) \ ?$$

Remark. Fujita (Acta Arith. **128** (2007), 349–375 ; Rocky Mountain J. Math. **39** (2009), 1907–1932) and He and Togbé (J. Number Theory **131** (2011), 120–137) obtained a positive answer for $D(-1)$-triples of the form $\{1, 2, c\}$, $\{F_{2k+1}, F_{2k+3}, F_{2k+5}\}$ and $\{1, k^2 + 1, k^2 + 2k + 2\}$, respectively.